

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY, CHANGA
M Sc (BIOCHEMISTRY)
SEMESTER II EXAMINATION
BC706: ENZYMOLOGY

Date: 04/05/2011

Time: 10.00 am to 1.00 pm

Total Marks: 70 minutes

Instructions to the candidates:

1. Attempt all questions
2. Questions in **Part A** should be answered in the question paper itself. Please do not write your candidate ID number on the question paper.
3. Write answers for Section I and Section II of **Part B** in separate answer sheets.

PART A

Duration: 30 minutes

Total marks: 20

Q1. Choose the correct option and put $\sqrt$ mark in front of it:

1. An enzyme with a nomenclature of 5.1.11.3 can be broadly classified as
(a) Ligase (b) Lyase (c) Isomerase (d) Transferase
2. An enzyme with its prosthetic group bound to it is known as
(a) Apo enzyme (b) Holo enzyme (c) Isoenzyme (d) Total enzyme
3. Which of the following does **NOT** occur during catalysis by chymotrypsin?
(a) Transition state stabilization (b) Covalent catalysis
(c) Electrostatic catalysis (d) None of the above
4. The catalysis by Ribonuclease A proceeds via which of the following mechanism?
(a) Covalent catalysis (b) Concerted acid-base catalysis
(c) Metal ion catalysis (d) catalysis by approximation
5. Which of the following is **TRUE** during enzyme purification?
(a) Specific activity increases, total protein decreases
(b) Specific activity decreases, total protein decreases
(c) Specific activity decreases, total protein increases
(d) Specific activity increases, total protein increases
6. Which of the following is the mode of regulation employed by lactate dehydrogenase?
(a) Allosteric activation (b) Occurrence as multienzyme complex
(c) Occurrence as zymogen (d) Presence of isozymes
7. Which of the following limits the rate of enzyme catalysis?
(a) V_{max} (b) K_M
(c) K_{cat} (d) k_{-1}
8. K_{cat}/k_M is apparent second-order rate constant of enzymatic reaction under the conditions of
(a) High substrate concentration (b) Low substrate concentration
(c) Both of the above (d) substrate concentration does not matter

9. The Lineweaver-Burk's plot analysis for a bi-substrate reaction shows parallel line. This indicates that the reaction is a
 (a) Ping-pong reaction (b) Sequential reaction
 (c) Ordered sequential reaction (d) Random sequential reaction
10. Kinetic data analysis based on Michaelis-Menten's equation is **NOT** applicable for
 (a) Uni-substrate reaction (b) Bi-substrate reactions
 (c) Allosteric reactions (d) None of the above
11. Which one of the following information related to mechanism of enzyme action can be determined using SDS PAGE?
 (a) Three dimensional structure of an enzyme (b) active site amino acids
 (c) Whether enzyme is oligomeric or monomer (d) binding site amino acids
12. Which of the following is **NOT** true for mixed inhibition?
 (a) Inhibitor can bind to both E and ES complex (b) Inhibitor binds at a site same as active site
 (c) V_{max} decreases (d) Non-specific inhibition is it's special case
13. Which of the following inhibitions is affected by substrate concentration?
 (a) Substrate inhibition (b) Competitive inhibition
 (c) None of the above (d) Both of the above
14. Binding of 2,3-BPG shifts the O_2 -saturation curve of Hb to the right. This indicates
 (a) O_2 dissociating ability of Hb increases (b) O_2 binding ability of Hb increases
 (c) O_2 dissociating ability of Hb decreases (d) O_2 binding ability of Hb cannot be judged from this information
15. Which of the following is suitable to describe positive co-operativity in Hemoglobin using Hill's Plot?
 (a) $n=1$ (b) $n<1$ (c) $n>1$ (d) n is not sufficient to predict the nature of co-operativity
16. Which of the following is **NOT** true for enzyme-substrate interaction?
 (a) Binding of substrate causes desolvation of active site
 (b) Distortion in substrate is higher than that in the enzyme
 (c) Conformational change is induced in the enzyme
 (d) None of the above

Q2. State True or False

- During immobilization, enzyme saturated with substrate is used to ensure that enzyme retains its correct active site orientation. [True] / [False]
- Organic molecules that are tightly bound to the enzymes and which participate in its catalytic reaction are known as prosthetic groups. [True] / [False]
- In multi-step reactions, the step with the highest transition state free energy is the rate determining step of the reaction. [True] / [False]
- Low K_i for an inhibitor I means, IC_{50} of I for that particular enzyme is high. [True] / [False]

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Instructions to the candidates:

1. Attempt all questions
2. Write answers for Section I and Section II of **Part B** in separate answer sheets.

Part B

Total Marks: 50

Section I

- Q.1 (A)** Explain various methods to study enzyme active site **5**
- OR**
- (A)** Explain the process of enzyme purification in general. Comment on the changes in parameters like enzyme activity, specific enzyme activity and total protein content after each purification step. **5**
- (B) Answer the following (ANY TWO)** **5**
- (i)** Substrate specificity of enzymes
 - (ii)** Compare enzymatic reactions vs chemical reactions
 - (iii)** Enzyme Nomenclature system and its importance
- Q.2 (A) Write short notes on (ANY TWO)** **4**
- (i)** Importance of binding energy in enzyme catalysis
 - (ii)** Induced Fit theory
 - (iii)** Catalytic triad of chymotrypsin active site
- (B)** Although chymotrypsin, trypsin and elastase, all are serine proteases, how do they have different specificity for protease action? **4**
- OR**
- (B)** Describe various mechanisms of enzyme catalysis. **4**
- (C)** Suggest a suitable experimental approach for deriving following information related to mechanism of enzyme action. **2**
- (i)** To assess the importance of particular amino acid present at the active site
 - (ii)** To know the three dimensional structure of an enzyme

Section II

- Q.3** (A) List various methods for determination of K_M and explain any one in detail 4
OR
(A) Explain how Lineweaver-Burk's plot can be used to distinguish between single displacement and double displacement kinetics. 4
(B) **Explain the following in brief (ANY TWO)** 6
(i) Catalytic efficiency of enzyme
(ii) Importance of K_M and V_{max} values
(iii) Mechanisms of bi-substrate reactions
- Q.4** (A) Explain the MNC model to explain the working of allosteric enzymes. 4
OR
(A) Give example and explain the advantages of multi enzyme complexes in metabolism. 4
(B) **Write a shortnote on of the following (ANY TWO)** 6
(i) Mechanism and kinetics of non-competitive inhibition
(ii) Mechanism of co-operativity in hemoglobin
(iii) Mechanisms of control and regulation of enzyme activity
- Q.5** (A) List various methods of enzyme immobilization and explain any one. 5
OR
(A) Describe biotechnological applications of enzymes 5
(B) Explain the following (ANY TWO) 5
(i) Factors influencing choice of carrier matrices for industrial purpose
(ii) Advantages of immobilized enzymes
(iii) Use of fluorescence in studying single molecule enzymology